1 ECE 215D Modern Micro-system (SOC)

We have sensors that convert from physical phenomena to electric signals; sensors exploit some physical phenomena to achieve this.

1.1 Signal Processing Sides

1.1.1 Recall Fourier transform and some key theorem

Time Domain $\leftrightarrow$ Frequency Domain It masures entire waveform/lifetime.

$$x(t) \leftrightarrow X(f)$$

Only real values $\leftrightarrow$ Always complex. It also means that $X(f)$ is hermitian will always be symmetric at $x = 0$.

$$\int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}dt \Leftrightarrow \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}df$$

Convolution

$$x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau)y(t-\tau)d\tau$$

Flipping and sifting: impulse train aka Π ("Shah")

$$x(t-T) \leftrightarrow X(f)e^{-j2\pi fT}$$

Shifting ("sampling") property of impulse function $\delta(t-T)$ based on fact:

$$\int_{-\infty}^{\infty} x(t)\delta(t-\tau)d\tau = x(\tau)$$

Theorems Convolution:

$$x(t) * y(t) \leftrightarrow X(f)Y(f) \text{ and vice versa}$$

Scaling: The a^{-1} accounts for the stretched area:

$$x(at) \leftrightarrow \frac{1}{|a|}X\left(\frac{f}{a}\right)$$

Shifting:

$$x(t-T) \leftrightarrow X(f)e^{-j2\pi fT}$$

Example applying the theorems

$$\cos(2\pi kt) \leftrightarrow \frac{1}{2}[\delta(f-k) + \delta(f+k)]$$

$$\cos(2\pi k_0 t) \leftrightarrow \frac{1}{2}[\delta(f-f_0) + \delta(f+f_0)]$$

Synthesizing a periodic waveform

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{j2\pi nt/T}$$

The spectrum is a plot of $x_p(t)$'s Fourier series.

$$x_p(t) = \sum_{n=-\infty}^{\infty} C_n e^{j2\pi nt/T}$$

By Euler identity $C_n = \text{Re}\times(C_n) + j\text{Im}\times(C_n)$

$$C_n = 2\left[\text{Re}(C_n)\cos\left(\frac{2\pi nt}{T}\right) + j\text{Im}(C_n)\sin\left(\frac{2\pi nt}{T}\right)\right]$$

1.2 Fourier Transform Tables

Property	Signal	Transform
Linearity	$\alpha x_1(t) + \beta x_2(t)$	$\alpha X_1(j\omega) + \beta X_2(j\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Conjugate symmetry	$x(t)$ real	$X^*(j\omega) = X(-j\omega)$ Magnitude: $\lvert X(-j\omega)\rvert = \lvert X(j\omega)\rvert$ Phase: $\angle X(-j\omega) = -\angle X(j\omega)$ Real Part: $X_r(-j\omega) = X_r(j\omega)$ Real Part: $X_j(-j\omega) = -X_j(j\omega)$
Conjugate antisymmetry	$x(t)$ imaginary	$X^*(j\omega) = -X(-j\omega)$ Magnitude: $\lvert X(-j\omega)\rvert = \lvert X(j\omega)\rvert$ Phase: $-\angle X(-j\omega) = \angle X(j\omega) \pm \pi$ Real Part: $X_r(-j\omega) = -X_r(j\omega)$ Real Part: $X_j(-j\omega) = X_j(j\omega)$
Real and Even	$x(-t) = x(t)$	$X(j\omega)$: Real and Even
Real and Odd	$x(-t) = -x(t)$	$X(j\omega)$: Imaginary and Odd
Real Even-Odd Decomposition	$x_e(t)$ $x_o(t)$	$Re(X(j\omega))$ $jIm(X(j\omega))$
Time Shifting	$x(t-T)$	$X(j\omega)e^{-j\omega T}$
Frequency Shifting	$x(t)e^{-j\omega_0 t}$	$X(j(\omega-\omega_0))$
Modulation Property	$x(t)\cos(\omega_0 t)$	$\frac{1}{2}[X(j(\omega-\omega_0)) + X(j(\omega+\omega_0))]$
Time and Frequency Scaling	$x(at)$, $x(a(t+T))$	$\frac{1}{\lvert\alpha\rvert}X(\frac{j\omega}{a})$, $\frac{1}{\lvert\alpha\rvert}X(\frac{j\omega}{a})e^{-j\omega T}$
Differentiation in Time	$\frac{d^n}{dt^n}(x(t))$	$(j\omega)^n X(j\omega)$
Differentiation in Frequency	$(-jt)^n(x(t))$	$\frac{d^n}{d\omega^n}(X(j\omega))$
Convolution	$x_1(t) * x_2(t)$	$X_1(j\omega)X_2(j\omega)$
Multiplication	$x_1(t)x_2(t)$	$\frac{1}{2\pi}X_1(j\omega) * X_2(j\omega)$
Integration	$\int_{-\infty}^{t} x(\tau)d\tau$	$\frac{X(j\omega)}{j\omega} + \pi X(0)\delta(\omega)$
Parseval's Theorem	$\int_{-\infty}^{\infty} \lvert x(t)\rvert^2 dt =$	$\frac{1}{2\pi}\int_{-\infty}^{\infty} \lvert X(j\omega)\rvert^2 d\omega$
Additional Properties	$x(t)$ Even and Real Odd and Real Even and Imaginary Odd and Imaginary	$X(j\omega)$ Even and Real Odd and Imaginary Even and Imaginary Odd and Real

Name	Signal	Transform
Rectangular Pulse	$x(t) = Arect(t/T)$	$X(j\omega) = AT\,\mathrm{sinc}\left(\frac{\omega T}{2\pi}\right)$
Triangular Pulse	$x(t) = A\Delta(t/T)$	$X(j\omega) = AT\,\mathrm{sinc}^2\left(\frac{\omega T}{2\pi}\right)$
Right-sided exponential	$x(t) = e^{(-at)}u(t)$	$X(j\omega) = \frac{1}{a+j\omega}$
Two-sided exponential	$x(t) = e^{(-a\lvert t\rvert)}$	$X(j\omega) = \frac{2a}{a^2+\omega^2}$
Signum Function	$x(t) = sgn(t)$	$X(j\omega) = \frac{2}{j\omega}$
Unit Impulse	$x(t) = \delta(t)$	$X(j\omega) = 1$
Sinc Function	$x(t) = \frac{\omega_0}{\pi}\frac{\sin(\omega_0 t)}{\omega_0 t}$	$X(j\omega) = rect(\frac{\omega}{2\omega_0})$
Constant-Amplitude Signal	$x(t) = 1,\ all\ t$	$X(j\omega) = 2\pi\delta(\omega)$
Inverse Function	$x(t) = \frac{1}{\pi t}$	$X(j\omega) = -j\ sgn(\omega)$
Unit Step Function	$x(t) = u(t)$	$X(j\omega) = \pi\delta(\omega) + \frac{1}{j\omega}$
Cosine	$x(t) = \cos(\omega_0 t)$	$X(j\omega) = \pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$
Sine	$x(t) = \sin(\omega_0 t)$	$X(j\omega) = \frac{\pi}{j}[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)]$
Duality	$x(t) \leftrightarrow X(j\omega)$	$X(t) \leftrightarrow 2\pi x(-\omega)$
Modulated Pulse	$x(t) = rect(t/T)\cos(\omega_0 t)$	$X(j\omega) = \frac{T}{2}\left[\mathrm{sinc}\left(\frac{(\omega-\omega_0)T}{2\pi}\right) + \mathrm{sinc}\left(\frac{(\omega+\omega_0)T}{2\pi}\right)\right]$
Note:	$\mathrm{sinc}(\alpha)$	$= \frac{\sin(\pi\alpha)}{\pi\alpha}$
	$rect(t/T)$	$= u(t+T/2) - u(t-T/2)$
IDTFT:	x(n)	$= \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega$

1.3 Z-Transform Tables

Property	Sequence	Transform	ROC
Linearity	$\alpha x_1[n] + \beta x_2[n]$	$\alpha X_1(z) + \beta X_2(z)$	At least the intersection of R_1&R_2
Time Shifting	$x[n - n_0]$	$z^{-n_0}X(z)$	R except for the possible addition or deletion of origin
Scaling in z	$e^{j\omega_0 n}x[n]$ $z_0^n x[n]$ $a^n x[n]$	$X(e^{j\omega_0}z)$ $X(z/z_0)$ $X(z/a)$	R $z_0 R$ $\lvert\alpha\rvert R$
Time Reversal	$\alpha x[-n]$	$\alpha X(z^{-1})$	R^{-1}
Time Expansion	$x_{(k)}[n] = \begin{cases} x[r], n = rk \\ 0, n \neq rk \end{cases}$	$X(z^k)$	$R^{1/k}$
Conjugation	$x[n]^*$	$X^*(z^*)$	R
Convolution	$x_1[n] * x_2[n]$	$X_1(z)X_2(z)$	Same as in Linearity
Accumulation	$\sum_{k=-\infty}^{n} = x[k]$	$\frac{1}{1-z^{-1}}X(z)$	Same as Linearity & $\lvert z\rvert > 1$
Differentiation in z	$nx[n]$	$-z\frac{dX(z)}{dz}$	R
Initial Value Theorem	if $x[n] = 0$ for $n < 0$	then	$x[0] = \lim_{z\to\infty} X(z)$

Z-Transforms

Signal	Transform	ROC
$x[n] = \sigma[n]$	$X(z) = 1$	All z
$x[n] = \sigma[n - m]$	$X(z) = z^{-m}$	All z except 0 (if $m > 0$) or ∞ (if $m < 0$)
$x[n] = \alpha^n u[n]$	$X(z) = \frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert > \lvert\alpha\rvert$
$x[n] = -\alpha^n u[-n-1]$	$X(z) = \frac{1}{1-\alpha z^{-1}}$	$\lvert z\rvert < \lvert\alpha\rvert$
$x[n] = n\alpha^n u[n]$	$X(z) = \frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\lvert z\rvert > \lvert\alpha\rvert$
$x[n] = -n\alpha^n u[-n-1]$	$X(z) = \frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2}$	$\lvert z\rvert < \lvert\alpha\rvert$
$x[n] = r^n \cos(\omega_0 n)u[n]$	$X(z) = \frac{1-[r\cos\omega_0]z^{-1}}{1-[2r\cos\omega_0]z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$
$x[n] = r^n \sin(\omega_0 n)u[n]$	$X(z) = \frac{r\sin\omega_0 z^{-1}}{1-[2r\cos\omega_0]z^{-1}+r^2z^{-2}}$	$\lvert z\rvert > r$